Indian Economy

3. Money

Introduction

Definition: Money is anything that is **accepted as a medium of exchange** for goods and services and for settling debts.

- Money is **not just currency notes and coins**.
- It includes anything people widely accept for payments—cash, bank deposits, and digital balances.

Characteristics of good money

- Acceptability: Everyone is willing to accept it.
- Durability: Does not wear out quickly.
- Portability: Easy to carry.
- Divisibility: Can be broken into smaller units.
- Uniformity: Every unit has the same value.
- Limited supply: Scarcity helps preserve value.

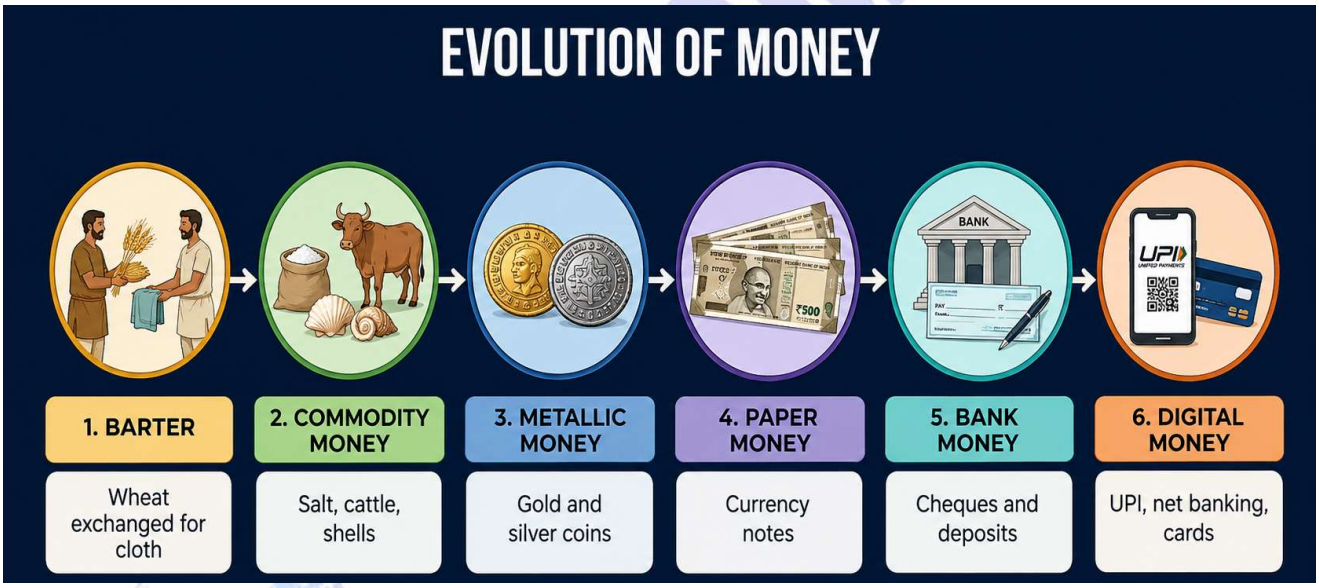
Four Core Functions of Money

1. **Medium of exchange:** Eliminates the need for barter. Example: You buy groceries using ₹500 instead of exchanging wheat for rice.
2. **Unit of account:** Provides a common measure of value. Example: A phone costs ₹20,000 and a laptop costs ₹60,000.
3. **Store of value:** Allows purchasing power to be saved for future use. Example: Keeping money in a bank account for later expenses.
4. **Standard of deferred payment:** Makes future payments possible. Example: Home loans and EMI payments are denominated in money.

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COMPARISON: INTRINSIC vs EXTRINSIC VALUE OF MONEY

ASPECT	INTRINSIC VALUE OF MONEY (Commodity / Metallic Money)	EXTRINSIC VALUE OF MONEY (Fiat / Paper / Digital Money)
BASIS OF VALUE	Intrinsic / Natural	Trust / Legally Enforced
SOURCE OF VALUE	Own Material Value	Government / Authority Backing
INHERENT WORTH	Present	Not Present
VALUE IF NOT USED AS MONEY	Retains Value	Nil Value
EXAMPLES	Gold, Silver, Salt, Cattle, Cowrie Shells	Currency Notes, Bank Deposits, Digital Money
ACCEPTABILITY	Limited (by Utility & Availability)	Wide (by Legal Tender & Trust)
DURABILITY	Generally Durable	Varies (Paper perishable, Digital depends on Technology)
DIVISIBILITY	Difficult	Easy
PORTABILITY	Difficult (Bulky / Heavy)	Easy
EXAMPLE IN CONTEXT	Metal Coins, Gold Bullion	Rupee Notes, Bank Balance, UPI



Barter System

- The barter system is a method of exchange in which goods and services are **directly exchanged for other goods and services** without using a medium of exchange.

Features

- No use of money
- Direct exchange of goods and services
- Value is decided through mutual agreement
- Common in early civilizations. Still seen in limited informal exchanges

Limitations:

- **Double coincidence of wants:** Both parties must want what other offers.

- No common measure of value
- Indivisibility of goods
- Difficulty in storing wealth
- Deferred payments are difficult

Commodity	Where it was used
Salt	Ancient Rome and Africa
Cattle	Early pastoral societies
Cowrie shells	India, China, and Africa
Tea bricks	Central Asia and Tibet
Tobacco	Colonial America

Commodity Money

Commodity money is a form of money in which **the item used as money has intrinsic value**, meaning it has value even outside its use as money.

Characteristics:

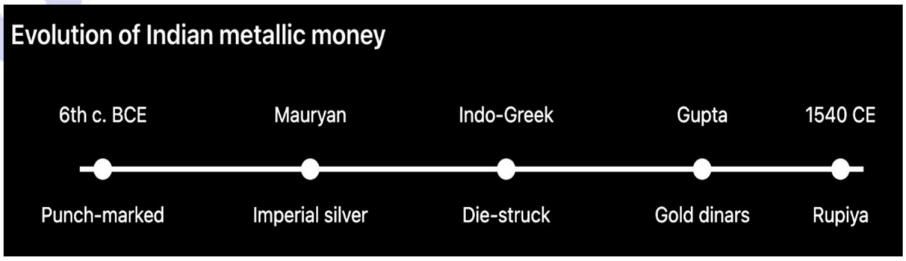
- Has intrinsic value. More acceptable than barter system
- Widely accepted within a community.
- Can be used for purposes other than exchange.
- Acts as a medium of exchange before metallic coins became common. Removes the need of double coincidence of wants.

Limitations:

- Lack of uniformity
- Poor divisibility
- Perishable
- Difficult transport
- Limited standardization

Metallic Money

- Metallic money is made from metals such as gold, silver, copper, bronze, or electrum, whose value is derived from the metal itself and, later, from the authority that minted it.
- It is durable, portable, divisible, and easier to standardize.
- The earliest metallic objects used for exchange existed in **Mesopotamia and Egypt**, but the first standardized coins are widely credited to the **Kingdom of Lydia.**



- **Benefits:** Metallic money offered intrinsic value, durability, portability, divisibility, standardization, and widespread acceptance, making trade more reliable and efficient.
- **Limitations:** Its limited supply, heavy transport, theft risk, wear and tear, coin debasement, and inability to support large-scale economic expansion eventually led to the adoption of paper money.

Gold Backed Paper Currency

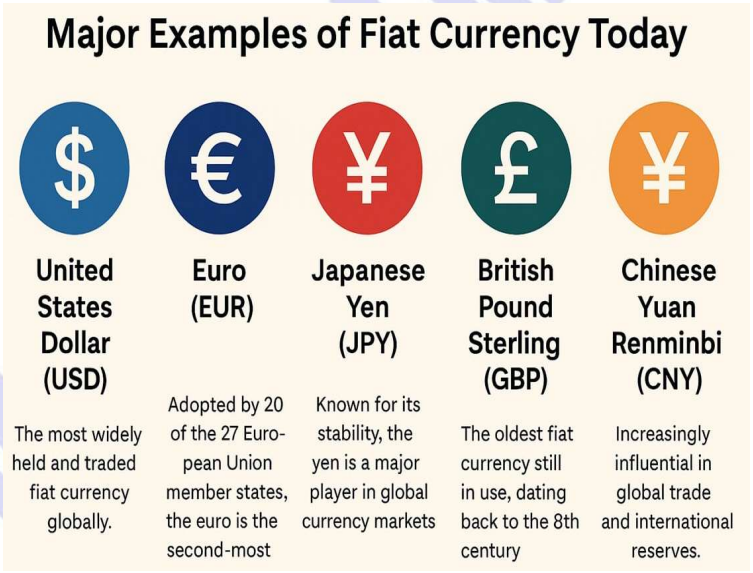
- Gold-backed paper currency was paper money that could be **exchanged for a fixed quantity of gold** on demand.
- The paper itself had little intrinsic value; its value came from the promise that the issuing authority would redeem it in gold.



  - **How it worked?** A central bank held gold reserves.
  - It issued paper notes.
  - Anyone could (in principle) exchange those notes for a fixed amount of gold.
  - This limited excessive money printing because every note required gold backing.

Fiat Currency

- It is money whose **value comes from government authority and public trust** rather than the intrinsic value of material from which it is made.
- **Characteristics of Fiat Money:**
  - No intrinsic value: Just paper or digital numbers.
  - Not backed by commodities: Value not tied to gold or silver.
  - Govt-issued: Central banks control production.
  - Trust-based: Value relies on public's faith in govt.
  - Easily printed: Govts can print more in need.
  - Highly regulated: Controlled by monetary policy.





- **Legal tender** is a form of money that a government officially recognises as acceptable for settling debts and transactions within its territory.
  - ➢ It must be **Fiat Money.**
- Bitcoin is **not a Fiat money,** yet it was legal tender in El-Salvador (till 2024). But it is an exception.
  - Legal to **own and use voluntarily**, but no longer compulsory legal tender in the practical sense after reforms in 2025.
- **Commemorative Coins =** Fiat money yes but may or may not be legal tender (Most of them are).
  - **Circulation commemorative coins:** Released into circulation and usable like ordinary coins.

- **Collector commemorative coins:** Sold in proof or UNC sets by the Government Mint and rarely used in everyday transactions.

- ₹100 silver commemorative coin can legally be used to pay ₹100, but collectors may pay several thousand rupees for it.
- We can use cheque / credit card to pay fees. So, why are they not fiat money/legal tender?

Limited vs Unlimited Legal Tender

- **Unlimited Legal Tender:**
  - Must be accepted for any amount (subject to the note being genuine and valid).
  - Banknotes are unlimited legal tender in India.
  - ₹10, ₹20, ₹50, ₹100, ₹200, ₹500 notes.
  - Reserve Bank of India Act, 1934

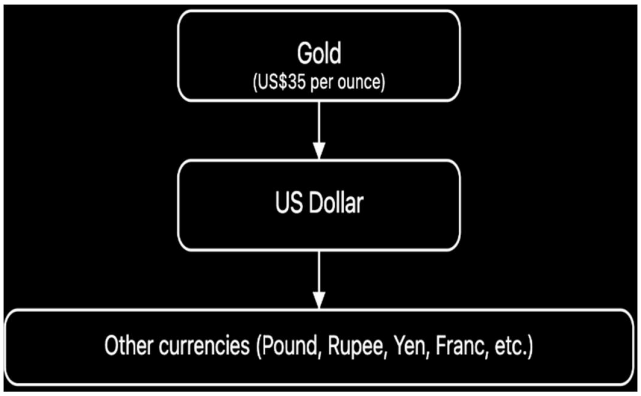
Current legal limits for coins (Coinage Act, 2011)

Coin	Legal Tender Limit
₹1, ₹2, ₹5, ₹10, ₹20 coins	Up to ₹1,000
50 paise coin	Up to ₹10
Coins below 50 paise	No longer legal tender (demonetized)

- **Limited Legal Tender:**
  - Must be accepted only up to a legally prescribed limit; beyond that, acceptance is voluntary.
  - Coinage Act, 2011

The Bretton Woods System (1944) and its End (1971)

- US promised to convert **US$35 into one troy ounce** of gold.
  - This promise applied mainly to foreign central banks
- The Dollar became the world's reserve currency.
- By the 1960s:
  - US spent heavily on Vietnam War and welfare programs.
  - More Dollars circulated worldwide than the gold available in US reserves.
  - France began demanding gold instead of Dollars.



- Economist **Robert Triffin** predicted a built-in contradiction.
  - The world needed more Dollars for international trade.
  - Supplying more Dollars weakened confidence in their gold convertibility.
- On 15 August 1971, President Richard Nixon ended the Dollar's convertibility into gold.

Petro Dollar System

- When President Richard Nixon ended gold convertibility, a major question arose:

- Why would the world continue holding dollars if they were no longer backed by gold?
- In 1973, the OPEC oil embargo caused oil prices to quadruple.
- The US recognized that if oil could be tied to the dollar, countries would continue needing dollars regardless of the end of gold backing.

1974 US–Saudi Understanding:

- **Saudi Arabia agreed to:** Price its oil in US dollars and Invest a large portion of its oil revenues in US Treasury securities and American financial markets.
- The United States agreed to: Provide military protection, supply weapons and offer security guarantees for the Saudi monarchy.

Evolution of the Indian Rupee

- The word Rupiya comes from the Sanskrit **Rupya,** meaning **wrought silver.**
- **Sher Shah Suri: (1540)**
  - Introduced the Rupiya made of silver.
  - Weight: about 178 grains (11.5 grams). High purity.
  - Uniform across the empire.
- **Mughal Standardization (1556–1707):**
  - Akbar expanded minting across the empire.
  - Hundreds of mints produced coins with standardized weight.
  - Coins carried the emperor's name.

Period	Nature of the Rupee
1540	Silver rupee introduced
Mughal era	Standard silver coin
1835	Uniform British Indian Rupee
1898	Gold Exchange Standard
1947	Independent India's currency
1957	Decimal currency
1971	Fiat currency
1993 onwards	Market-determined exchange rate

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- **East India Company Rupee (1757–1835)**
  - The Company initially continued Mughal-style coinage.
  - The three Presidencies (Bombay, Calcutta, Madras) minted different coins.

Coin	Metal
Mohur	Gold
Rupee	Silver
Dam	Copper

- **Currency Act of 1835:**
  - One uniform Indian Rupee with standard design and Queen/King's portrait
  - Single legal tender across British India
- **Paper Currency Act, 1861:** The British government obtained the exclusive right to issue banknotes.
- **Silver Crisis and Gold Exchange Standard (1873–1947):**
  - In the late nineteenth century: Large silver discoveries occurred and Silver prices collapsed.
- **The Fowler Committee (1898):**
  - The committee recommended: Linking the rupee to gold.
  - India adopted a Gold Exchange Standard.
- **Reserve Bank of India (1935):**
  - The Reserve Bank began issuing currency. Initially, notes carried: King George V and later King George VI

Decimalization (1957)

Old system
1 Rupee = 16 Annas = 64 Paise = 192 Pies

New system
1 Rupee = 100 Paise

- **Independent India's Rupee (1947–1957):**
  - King's portrait disappeared and Ashoka Lion Capital replaced imperial symbols.
  - The rupee remained linked to the British monetary system for a while.
- **Decimalization (1957):** Anna System was aborted via the **Indian Coinage (Amendment) Act, 1955.**
- **1971: End of Gold Convertibility**
  - The global monetary system changed.
  - India gradually moved toward a fiat currency.
- **1993:** India adopted a **market-determined managed float.**
  - The RBI now intervenes mainly to smooth excessive volatility rather than fix the exchange rate.
- 2010: The ₹ symbol was officially adopted.
  - Designed by D. Udaya Kumar using elements of both Devanagari and Roman scripts.
- **2022: Digital Rupee: A legal tender digital currency** denominated in Indian rupees.
  - October 2022: Wholesale pilot (e₹-W) for interbank settlements.
  - December 2022: Retail pilot (e₹-R) for the public.
  - The RBI has gradually expanded participating banks, merchants, and users.

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Indian Economy

MUHAMMAD BIN TUGHLAQ'S

TOKEN CURRENCY EXPERIMENT

(1329–1330 CE)

An ambitious attempt by Sultan Muhammad bin Tughlaq, ruler of the Delhi Sultanate, to introduce brass and copper coins as token currency equal to silver coins.

Brass / Copper Token Coin = Silver Tanka (Equal Value)

BACKGROUND	WHAT WAS THE EXPERIMENT?	WHY DID HE INTRODUCE IT?	WHY DID IT FAIL?
• Heavy military expenditure (wars in Deccan & Central Asia) • Shortage of silver in the treasury • Need to increase money supply for administration & trade	Around 1329–1330 CE, he issued brass and copper coins and declared them equal in value to the existing silver Tanka. **SILVER TANKA** Intrinsically valuable (made of silver) **=** **BRASS & COPPER TOKEN COINS** Value based on the royal order (face value)	• Reduce dependence on scarce silver • Increase money supply in the economy • Finance wars and administration • Facilitate trade with cheaper coins	The experiment collapsed due to loss of public trust and lack of effective state control. • **WIDESPREAD COUNTERFEITING** Copper coins were easy to imitate; fake coins flooded the market. • **NO SECURE MINTING TECHNOLOGY** No proper system to distinguish genuine coins from counterfeit ones. • **WEAK ADMINISTRATIVE ENFORCEMENT** State could not stop the minting and circulation of fake coins. • **PUBLIC CONFIDENCE COLLAPSED** Traders and people refused to accept the new coins.

HOW DID TUGHLAQ RESPOND?	ECONOMIC SIGNIFICANCE	COMPARISON WITH MODERN FIAT CURRENCY
He withdrew the token currency and allowed people to exchange copper coins for silver coins. This led to enormous losses to the royal treasury.	The experiment showed that money derives value not only from the metal it is made of but also from trust, credibility and effective state regulation.	**TUGHLAQ'S TOKEN CURRENCY** • Copper & brass coins • State-declared value • Weak anti-counterfeit measures • Failed due to loss of trust **MODERN FIAT CURRENCY** • Paper and digital money • State and Central Bank backing • Advanced security features • Works due to institutional credibility and public trust

Indian Economy

THE JOURNE

Silver Tanka (Established Currency) ➔ Brass & Copper Coins Introduced as Token Currency (1329–1330 CE) ➔ Counterfeiting & Loss of Trust ➔ Withdrawal & Huge Financial Loss

KEY TAKEAWAY

Public trust and effective regulation are the foundation of any successful currency.

Features of the Indian Notes

- **Accessibility Features:** Designed for visually impaired users.
  - Raised printing, Identification marks, Larger numerals, Distinct note sizes.
- **₹1 Note:**
  - Issued by the Govt of India
  - Contains the signature of Finance Secretary of India
  - RBI manages the circulation of this note.
- **https://paisaboltahai.rbi.org.in/:** You can access the features of our notes from this website of RBI.
- **Languages:**
  - **15 languages** appear in the language panel on the reverse.
  - Along with Hindi (front) and English (front), the note represents 17 languages in total.
- **Major security features:**
  - Watermark of Mahatma Gandhi.
  - Security Thread with "भारत", "RBI", and denomination.
  - Latent Image visible when tilted.
  - See-through Register that forms the denomination when viewed against light.
  - Micro-lettering reading "RBI" and the denomination.
  - Optically Variable Ink on higher denominations.
  - Intaglio Printing (raised printing) for tactile feel.
  - Number Panels with increasing numeral size.

Promise on the Note

- It is known as the **Promise Clause,** and it signifies that the Reserve Bank of India (RBI) guarantees the face value of the note.
- **Bearer** means that whoever physically possesses the note can use it.
- **Colonial era:** Promised to convert into full bodied gold or silver bars / coins worth the equal value in weight.
  - Same was with the Dollar during the **Bretton Woods System** and the Pound Sterling during the British Rule.



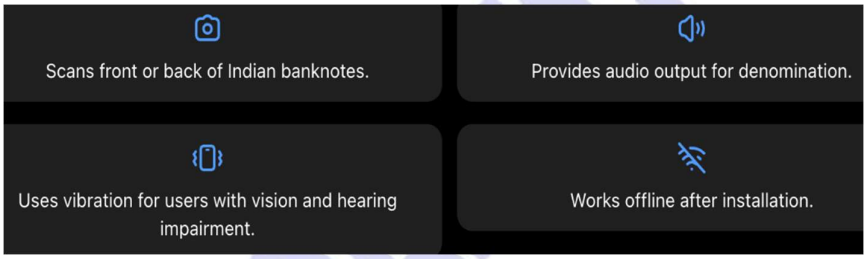
- **Modern Currency:**
  - Not inflation adjusted value. Not linked with weight of gold or silver.
  - Currency note is zero interest, anonymous bearer bond.

Indian Economy

- An Indian currency note **is not a promissory note** in the legal sense, even though it contains a promise clause.
    - **Section 4 of the Negotiable Instruments Act, 1881:** A promissory note is: "An instrument in writing (not being a bank-note or a currency-note) containing an unconditional undertaking.

MANI App

- **MANI (Mobile Aided Note Identifier):** It is a free mobile app launched by the Reserve Bank of India on 1 January 2020.
- To help visually impaired persons identify the denomination of Indian banknotes independently.
- Android and iOS both.
- Works offline after installation
- Supports 15 Languages, including Hindi and English



Demonetization

- Demonetization means the REMOVAL of legal tender status of currency notes. It becomes invalid for transactions.
- India has carried out three major demonetisations—in **1946, 1978, and 2016**.
    - **8 November 2016:** PM Modi announced that existing: ₹500 notes and ₹1,000 notes would cease to be legal tender from midnight.
- A **withdrawal of ₹2,000 notes in 2023**, which was not a demonetisation because the notes remained legal tender.
    - 2016-Nov: RBI introduced ₹2000 note, to meet the currency requirement of the economy in an faster manner after demonetisation of ₹500 and ₹1000
    - 2018–19: printing of ₹2000 banknotes was stopped.
    - 2023-May-23rd: RBI ordered people to get it exchanged with other notes.

Year	Government	Currency affected
1946	British India	₹500, ₹1,000, ₹10,000
1978	Morarji Desai	₹1,000, ₹5,000, ₹10,000
2016	Narendra Modi	₹500, ₹1,000
2023	Narendra Modi	₹2,000 withdrawn (not demonetised)

Gresham's Law

- It states that **"bad money drives out good money"** when both are accepted as legal tender at the same face value.
- **It requires:**
    - Both currencies are legal tender.
    - The government fixes their exchange value.
    - One currency is undervalued relative to its intrinsic value.

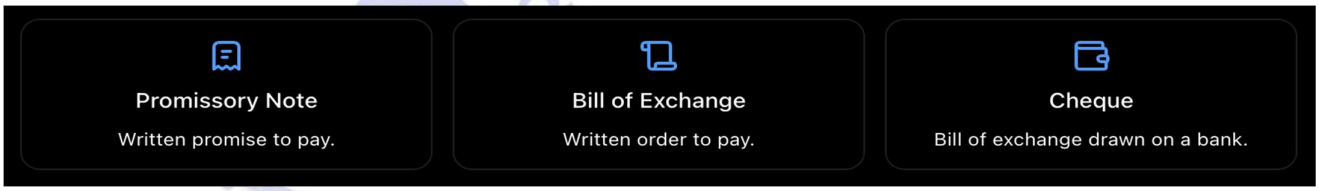
- People are free to choose which one to spend.

Good Money	➔ Everyday circulation ➔	Bad Money
High intrinsic value People hoard or melt it		Lower intrinsic value People spend it

- England in the 16th century provides the classic example. During the reign of Henry VIII, many silver coins were debased by reducing their silver content.
- Since old and new coins had the same face value, people hoarded the older, purer coins, leaving mostly debased coins in circulation.

Bank Money / Deposit Money

- It is the **money that exists as deposits in bank accounts** and is **transferred through banking instruments** rather than as physical cash.
- It forms the largest part of the modern money supply.
- Payments can be made through cheques, UPI, debit cards, NEFT, RTGS, and other banking channels.
- **Negotiable Instruments Act, 1881 (NIA):**
  - A negotiable instrument is a written document that **promises or orders payment of a specified sum** and can be transferred from one person to another.
  - Provides legal framework for **cheques, bills of exchange, promissory notes,** making them transferable (negotiable) and legally enforceable.
  - 3 instruments are covered under this Act:



Negotiable Instruments

1. **Promissory Note**
   - A promissory note is a written promise by one person to pay another person a certain amount.
   - Example: "I promise to pay Rahul ₹50,000 on 1 January."
   - Parties involved: Maker and Payee
2. **Bill of Exchange:**
   - A bill of exchange is a written order directing someone to pay another person.
   - **Example:** A wholesaler orders a retailer to pay ₹1 lakh after 90 days.
   - Parties involved: Drawer (wholesaler), Drawee (retailer), Payee (Wholesaler or any other person named by him).
3. **Cheque:**

- A cheque is a special type of bill of exchange drawn on a bank and payable on demand.
- Parties involved: Drawer (account holder), Drawee (bank), Payee
- **Section 138: Cheque dishonour:** A cheque is dishonoured for reasons such as insufficient funds, and the drawer does not make payment after receiving the statutory demand notice.
  - ➢ Makes cheque bouncing a criminal offence

Demand Draft and Overdraft Facilities

Demand Draft

- It is a prepaid payment instrument issued by a bank
- To transfer money securely.
- The payment is guaranteed by the bank because the customer pays the amount to the bank before the draft is issued.



Overdraft Facility:

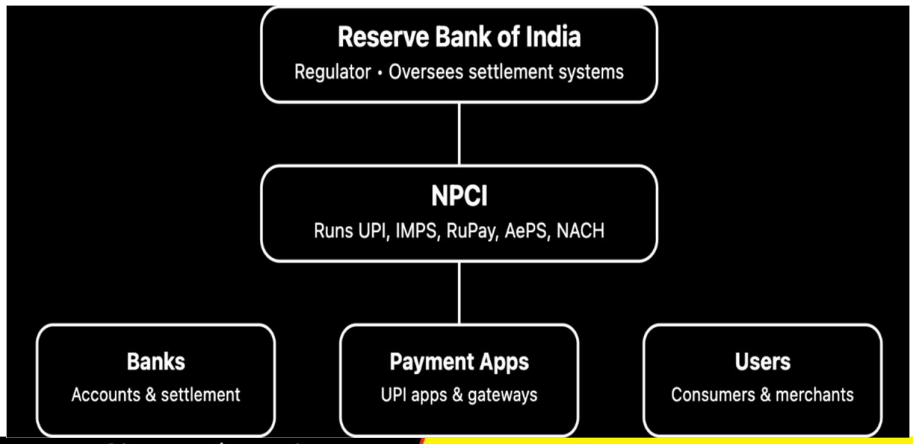
- It is a short-term credit facility that allows a customer to withdraw more money than is available in their bank account.
- Under the **Pradhan Mantri Jan Dhan Yojana,** eligible account holders may receive an overdraft facility

Banking Codes

- **MICR code:**
  - Magnetic Ink Character Recognition.
  - 9 digits code written in Iron Oxide ink for automated clearance.
  - It allows cheque-sorting machines to read and process cheques quickly and accurately.
- **Indian Financial System Code (IFSC):**
  - It is an 11-character alphanumeric code that uniquely identifies every bank branch participating in electronic payment systems.
  - IFSC is required for: NEFT, RTGS, and IMPS payments.
  - It ensures that money reaches the correct bank branch electronically.

Digital Payments

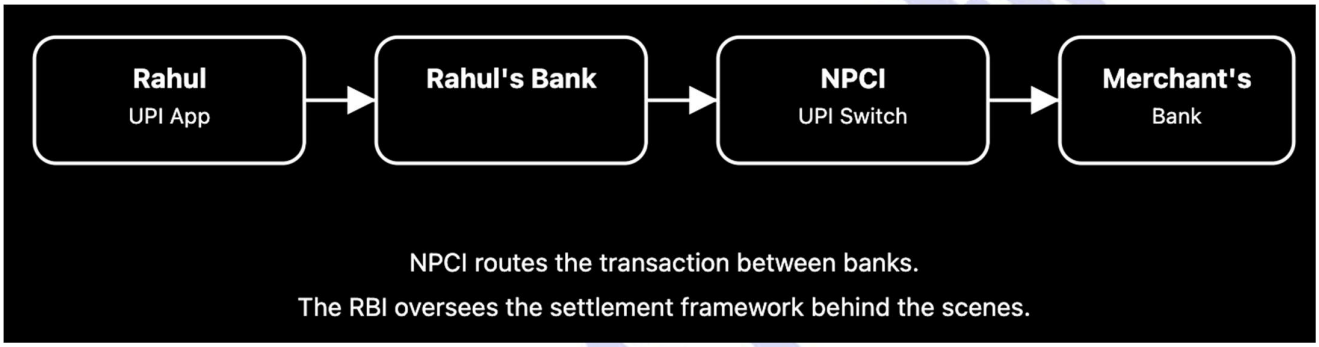
- RBI is the apex authority for payment and settlement systems under the **Payment and Settlement**



Systems Act, 2007.

- NPCI is a **not-for-profit organisation** established in 2008 by the RBI and the Indian Banks' Association.
  - Its job is to build and operate retail payment infrastructure.
  - NPCI does not hold customer deposits—it provides the infrastructure connecting banks.
- **Banks and Payment Service Providers:**
  - The third layer consists of: commercial banks, payment banks, small finance banks, payment apps, payment gateways, and merchants.
  - They provide the customer-facing services.

How a UPI payment actually works?



The process happens in seconds:

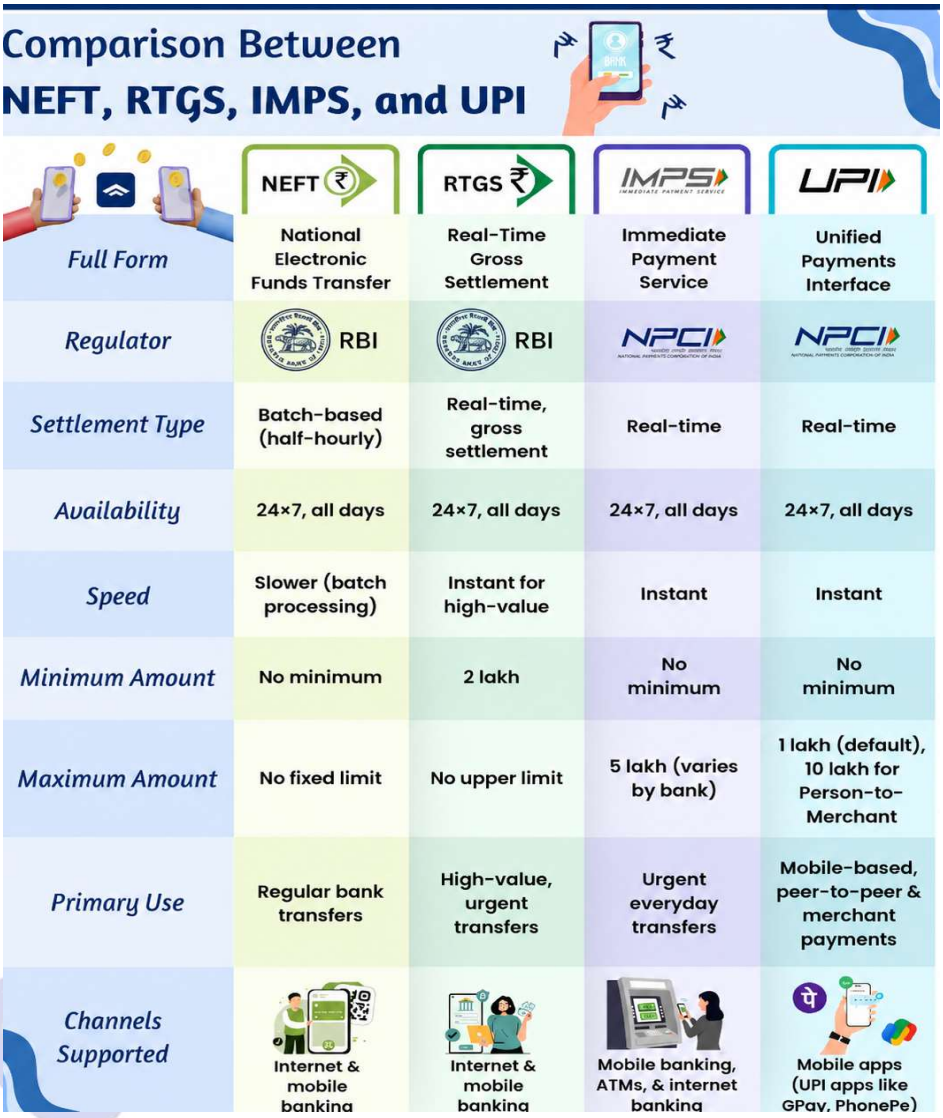
- Rahul enters his UPI PIN.
- The UPI app sends the request.
- NPCI's UPI switch identifies both banks.
- Rahul's bank verifies funds.
- Merchant's bank receives confirmation.
- Settlement between banks occurs through the RBI's payment infrastructure.

UPI Push and Pull Transactions

- **Push Transactions:**
  - When you send money by your own discretion
  - E.g. sending money to a relative/friend or making a payment to a tea seller.
- **Pull transaction:**
  - When you have given command that every month electricity company should cut the money from my bank account
  - The company will pull money without requiring you to push/click buttons every month.
- UPI Delegated Payments to create Secondary Accounts (2024-Aug)
  - Father (primary account-UPI) creates a secondary account-UPI daughter.
  - Then she can use money from father's bank account upto Rs.X limit using her mobile.

Other Digital Payment Systems

1. **NEFT (National Electronic Funds Transfer):**
   - Available 24×7. Free of cost
   - Transfers occur in half-hourly batches.
   - Suitable for ordinary bank transfers.
   - Mainly for small amounts upto 5 lakh (Limit varies from bank to bank)



2. **RTGS (Real Time Gross Settlement):**
   - Real-time settlement. Instant Process. Available 24×7.
   - No batching.
   - Primarily used for high-value transactions.
   - Operated directly by RBI.
   - RBI allowed bankers to charge upto ₹50 fees from the sender.
3. **IMPS (Immediate Payment Service):**
   - ₹1 to maximum ₹5 lakhs
   - Immediate transfer.
   - Available round the clock.
   - Works through mobile and internet banking.

AEPS and BBPS

1. **AEPS (Aadhaar Enabled Payment System)**

- Enables banking transactions using **Aadhaar and biometrics,** primarily for financial inclusion.
- Authentication and payment is done via Bank Mitras.
- Helps in Direct Benefits Transfer, helping government subsidies reach the grassroots levels.

2. **BBPS (Bharat Bill Payment System)**
   - Provides a single interoperable platform for paying bills such as electricity, mobile, gas, insurance, etc.

NPCI manages both AEPS and BBPS.

Rupay Card Payment Gateway (2012)

- **Rupee + Payment = RuPay card .** It was launched by NPCI.
- It is world's **7th payment gateway** similar to Mastercard, Visacard, China's Union Pay.
- Rupay **charges less fees** than other companies and is India's domestic payment system.
- **Types of Cards:** RuPay debit, credit, and prepaid cards.
- Rupay debit card **given free with PMJDY** (Jan Dhan) Bank account.
- One of India's major payment innovations is **linking RuPay Credit Cards to UPI**.



E-Rupi by NPCI (2021)

- E-Rupi is an SMS /QR Code based **Prepaid Cashless Electronic Voucher** created by **National Payment Corporation of India (NPCI)** using its UPI Platform.
- It is not a payment app, bank transfer, or digital currency.
- Instead, it is a **person-specific and purpose-specific voucher** that can only be used by the intended beneficiary for the intended service.
- Think of it as a **digital gift coupon** with restrictions.
  - Suppose the government wants to pay for your COVID vaccination costing ₹1,000.
  - Instead of transferring ₹1,000 into your bank account, it sends you an e-RUPI voucher worth ₹1,000 via SMS or QR code.
  - You can redeem it only for vaccination at an authorized center—not withdraw cash or spend it elsewhere.

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Merchant Discount Rate

- MDR is the fees that a merchant must pay to his (acquirer) bank for every card transaction.
- The merchant does not pay the entire MDR to one entity. It is shared among:
  - Acquiring Bank (merchant's bank)
  - Issuing Bank (customer's bank)
  - Card Network (RuPay, Visa, Mastercard, etc.)
  - Payment Service Provider/Gateway, where applicable
- The merchant pays MDR, not the customer.



Project Nexus

- It is a **multilateral cross-border payments initiative** led by the Bank for International Settlements (BIS) Innovation Hub that aims to connect the **fast payment systems (FPS) of different countries**,
- **Objective: A**llowing people to send money internationally as easily as making a domestic **UPI payment.**
- India's UPI is one of the **founding payment systems** participating in this project.
- Today, sending money abroad usually involves multiple correspondent banks, higher fees, and delays.

Country	Fast Payment System
India	UPI
Singapore	PayNow
Malaysia	DuitNow
Philippines	InstaPay
Thailand	PromptPay

- Project Nexus creates a **single standardized connection between national instant payment systems**, so countries do not need to build separate payment links with every other country.
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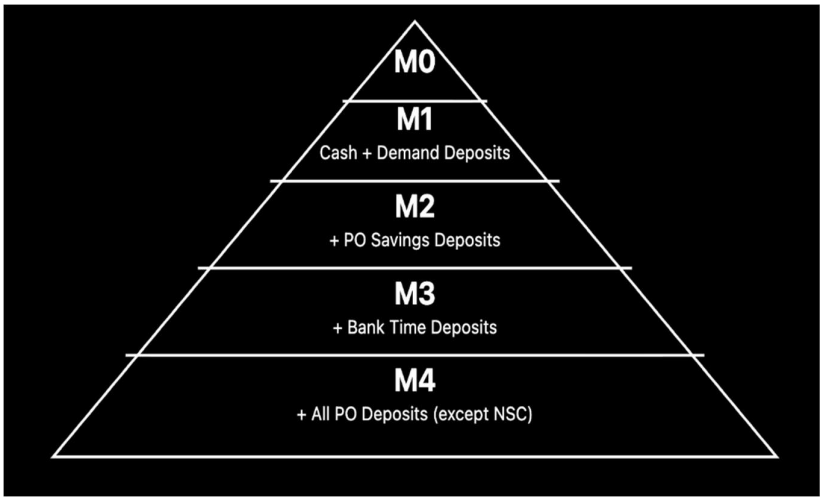
Plastic Currency Notes by the RBI

- These are the currency notes made from security-grade **Biaxially Oriented Polypropylene (BOPP)** instead of the traditional cotton-based paper substrate.
- They are **specially engineered polymer banknotes** with embedded security features, not ordinary plastic.
- RBI will start with **small denomination notes** such as ₹10 and ₹20. Up to 4x longer time span.
- **2013:** Government approved a field trial of ₹10 polymer notes.
- **Existing paper notes will continue** to circulate alongside polymer notes; this is not a currency replacement exercise.
- The main objective is to make Indian currency **last longer, become harder to counterfeit, and reduce replacement costs.**
- More than **50 countries** use polymer banknotes. **Australia** was the 1st country to introduce them in **1988.**

Crypto Currency

- It is a digital / virtual currency created & stored using blockchain technology.
- It is a privately created digital asset. It is **not legal tender in India** and is not issued by the RBI.
- It relies on a distributed ledger to **record ownership** without a central authority like a bank or government.
- **Why Cryptocurrency?**
    - Some anarchist groups lost faith in FIAT MONEY because **USA's Subprime Crisis (2007)** eroded the purchasing power of US Dollar.
    - They also **dislike Banks & Card Companies** because of transaction charges on e-banking, card payments, MDR, interoperability issues.

Money Supply



- Money supply is the **total amount of money** available in an economy at a given point in time that people and businesses can use for spending, saving, or making payments.
- The Reserve Bank of India (RBI) measures money supply using M0, M1, M2, and M3, with M3 (Broad Money), M4 being the most important indicator for monetary policy.

- In an economy, money circulates continuously between households, businesses, banks, and the RBI.
- **M0:** M0 is the **money created directly** by the RBI.
    - It is called Reserve Money or High-Powered Money because it forms the base from which banks create additional money through lending.
- **M0 = C + OD + Bankers Deposits**
    - C = Currency in circulation
    - OD = Other deposits with RBI
    - Bankers' Deposits = Commercial banks' deposits with RBI (including CRR balances)
- When banks receive reserves from the RBI, **they can lend multiple times**, expanding the money supply.
- **M1 – Narrow Money**
    - M1 is the **most liquid money** available to the public for everyday transactions.
- **M1= Currency + Demand Deposits + Other Deposits with RBI**
- It includes:
    - Currency with the public
    - Demand deposits (current and savings accounts that can be withdrawn on demand)
    - Other deposits with RBI
- It is **called Narrow Money** because every component can be spent almost immediately.
- **M2 – Intermediate Money**
    - M2 is a broader measure than M1.
- M2 = M1 + Savings Deposits with Post Office Savings Banks.
- It adds **post office Savings Bank deposits** (excluding National Savings Certificates and similar instruments).
- These deposits are **fairly liquid but slightly less convenient** than ordinary bank deposits.
- **M3 – Broad Money**
    - It is the most important money supply measure in India.
    - When newspapers report money supply growth, they usually mean M3.
- M3 = M1 + Time Deposits with Banks
- It includes:
    - Everything in M1
    - Fixed deposits
    - Recurring deposits
    - Other time deposits with commercial banks
- The RBI watches M3 because it helps indicate:
    - Inflationary pressures
    - Liquidity in the economy
    - Credit growth
    - Effectiveness of monetary policy
- **M4:**
    - It is the broadest measure of money supply in India.

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- It includes almost all liquid money available in the economy, including Post Office deposits that are not included in M3.

- M4 = M3 + All Deposits with Post Office Savings Banks (excluding National Savings Certificates)
- The National Savings Certificate (NSC) is not counted in M4 because:
  - It is a long-term savings instrument, not readily spendable money.

Practice Questions

Find correct statement(s) about "Blockchain Technology" [UPSC-Prelims-2020]

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. Structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that use basic features of blockchain can be developed without anybody's permission.

Answer Codes:

a) 1 only
b) 1 and 2 only
c) 2 only
d) 1 and 3 only

Find correct statements about RTGS and NEFT: (Pre'25)

1. In RTGS, the settlement time is instantaneous while in case of NEFT, it takes some time.
2. In RTGS, the customer is charged for inward transactions while that is not the case for NEFT.
3. Operating hours for RTGS are restricted on certain days while this is not true for NEFT.

Codes:

(a) I only
(b) I and II
(c) I and III
(d) III only

Find Correct Statement(s) [Prelims-2018]

1. BHIM app allows the user to transfer money to anyone with a UPI-enabled bank account.
2. While a chip-pin debit card has four factors of authentication, BHIM app has only two factors of authentication.

Answer codes:

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Find the correct statement(s) [Prelims-2017]

1. NPCI helps in promoting the financial inclusion in the country.

2. NPCI has launched RuPay, a card payment scheme.

Answer codes:

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

What is "Merchant Discount Rate" ? (IAS-Prelims 2018)

A. The incentive given by a bank to a merchant for accepting payments through debit cards pertaining to that bank.

B. The amount paid back by banks to their customers when they use debit cards for financial transactions for purchasing goods or services.

C. The charge to a merchant by a bank for accepting payments from his customers through the bank's cards.

D. The incentive given by the Government, to merchants for promoting digital payments by their customers through Point of Sale (PoS) machines and debit cards.

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